



SIMULATION AND OPTIMISATION OF MICROGRID: ELECTRICITY AND HEAT SUPPLY

Group 02

Simulation and Optimisation of Microgrid: Electricity and Heat supply

01

Introduction

02

System component sizing and Selection

03

Simulation and Optimization of Energy model

04

Result and Analysis

05

Economical calculation

06

Ecological comparison

07

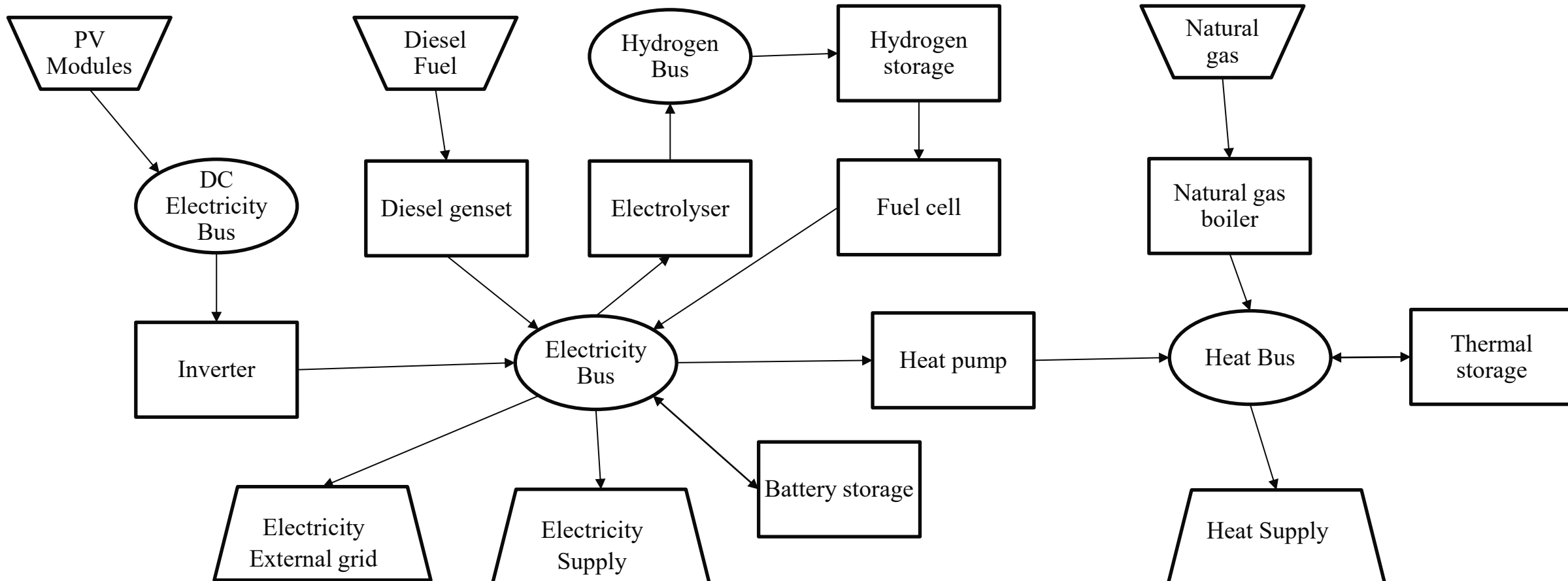
Conclusion and Recommendation

08

References

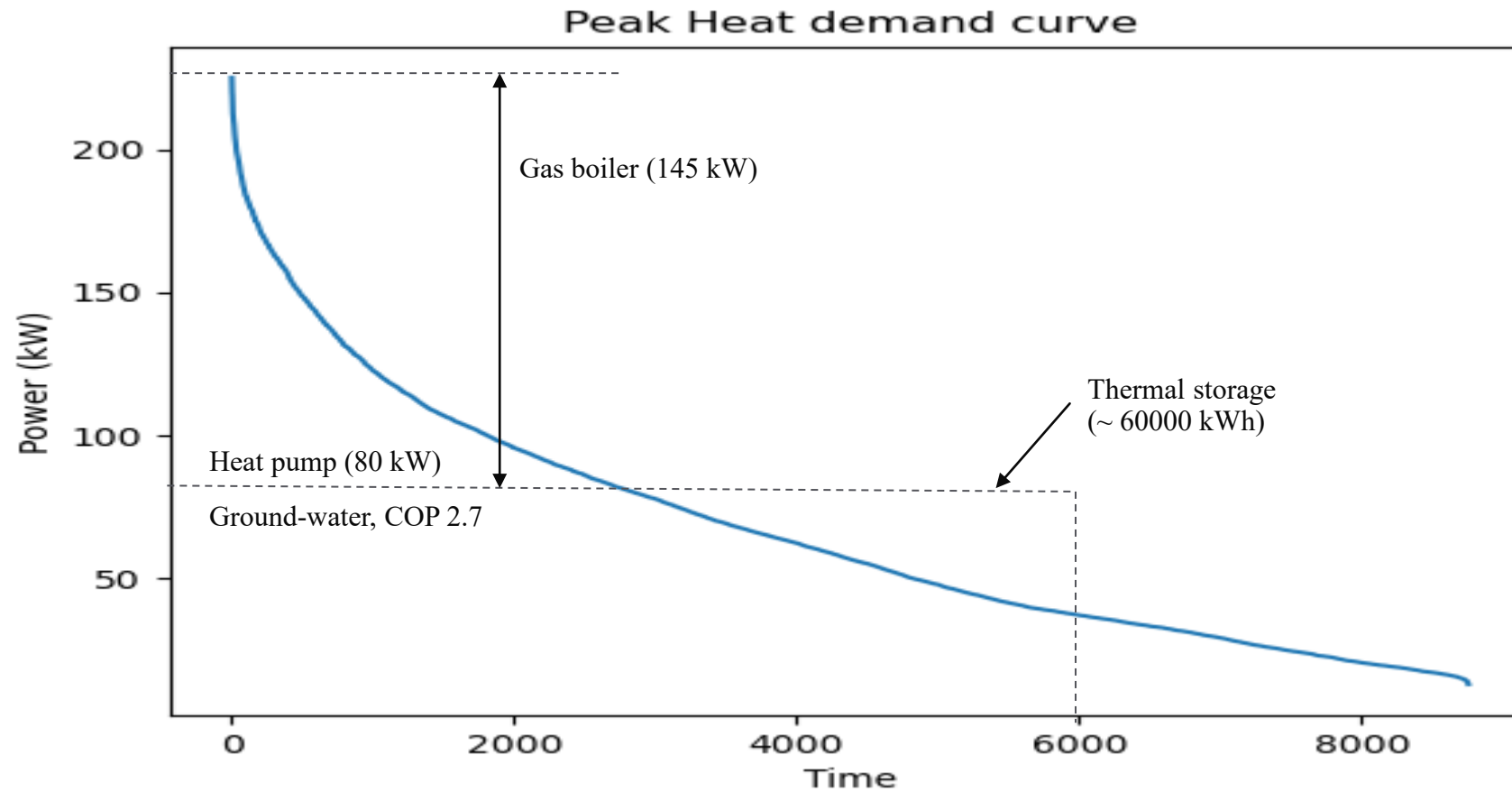
1. Introduction

- **Task : To design a microgrid for the Electricity and Heat supply of a small town and optimization of microgrid using python.**
- **How the Electricity and Heat should be supplied?**



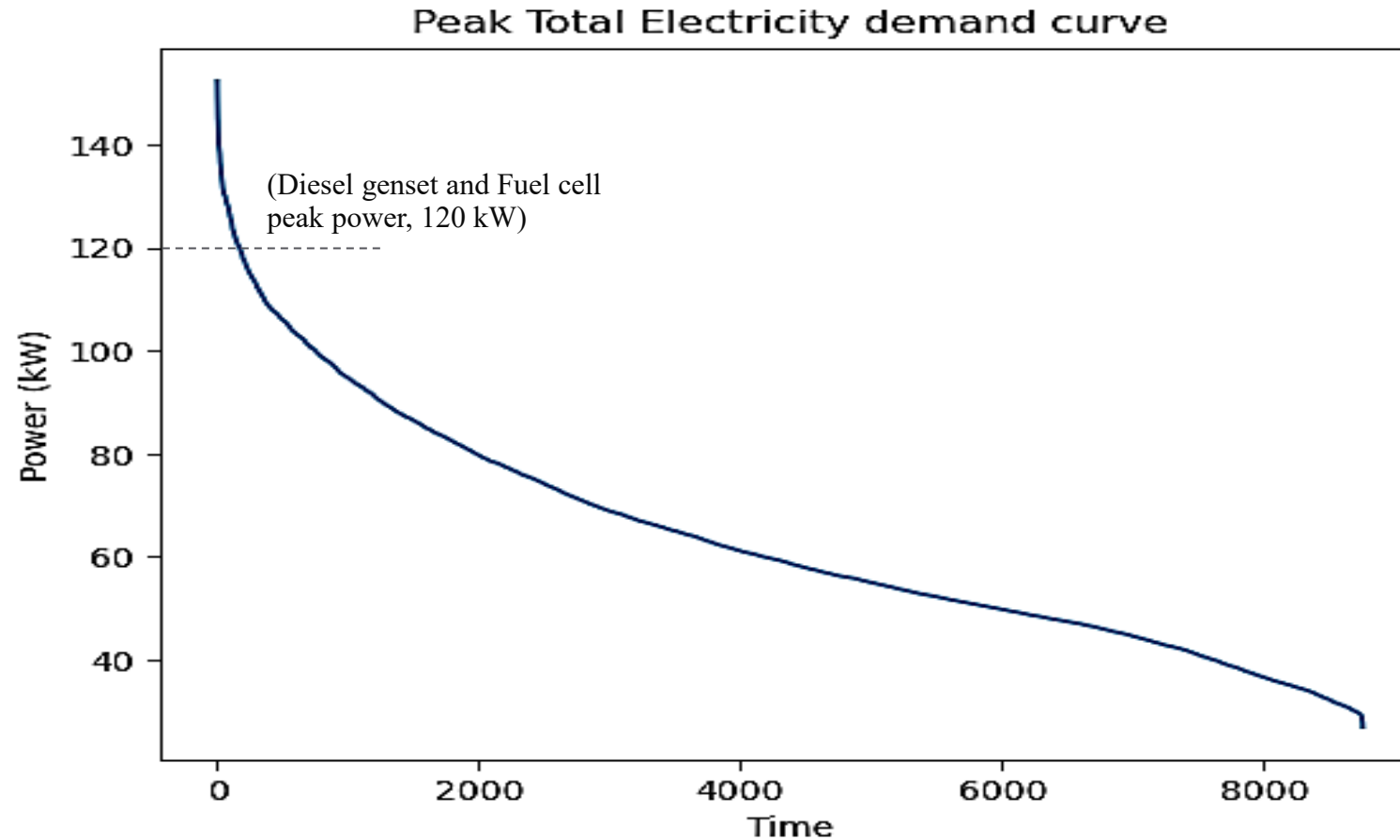
2. System component sizing and Selection

- Heating component sizing



2. System component sizing and Selection

- Electrical component sizing :



2. System component sizing and Selection

- **Electrical component sizing :**
 - ❑ **Battery storage :** Capacity = 1000 kWh, Daily energy storage, Based on average daily electricity consumption
 - ❑ **Electrolyser :** Input power capacity = 120 kW
 - ❑ **Hydrogen storage :** Flexible, Based on amount of hydrogen produced
 - ❑ **PV system :** Total possibly installed capacity based on available land and size of selected PV module,
PV roof area = 1200 m², Ground area = 0.75 ha = 7500 m²
 - ❑ **Inverter :** Based on PV system capacity

Note : The size of the energy model component can be modified based on the optimisation result. For hydrogen production process water purification and battery module has been not considered while the hydrogen loss during drying process calculated as hydrogen battery charging loss.

2. System component sizing and Selection

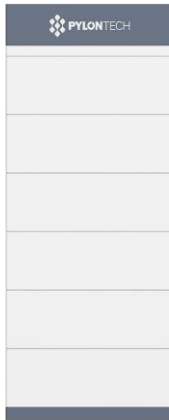
- **PV modules and Inverter selection:**



- Model : JA Solar, JAM54D41-450-LB
- Type of panel : Bifacial monocrystalline
- Peak power: 450 W
- Area per module $\sim 2 \text{ m}^2$
- Possibly installed capacity = 1470 kW (Min. space between installation of plate considered)
- Total modules installed: 3267 modules
- Life span : 25 Year
- **Inverter** : Solis, S6-EH3P10K-H-EU
- Capacity : 1500 kW
- Efficiency : 98.2%
- Life span : 10 Years

2. System component sizing and Selection

- **Battery storage:**



- Model : Polyntech force H3
- Capacity per module : 35.84 kWh
- No. of modules : 28 modules
- Charging/ Discharging power ~ 144 kW
- DOD = 95%
- Charging/Discharging efficiency = 98%
- Self discharge rate = 0.003% per day
- Life span = 15 Years

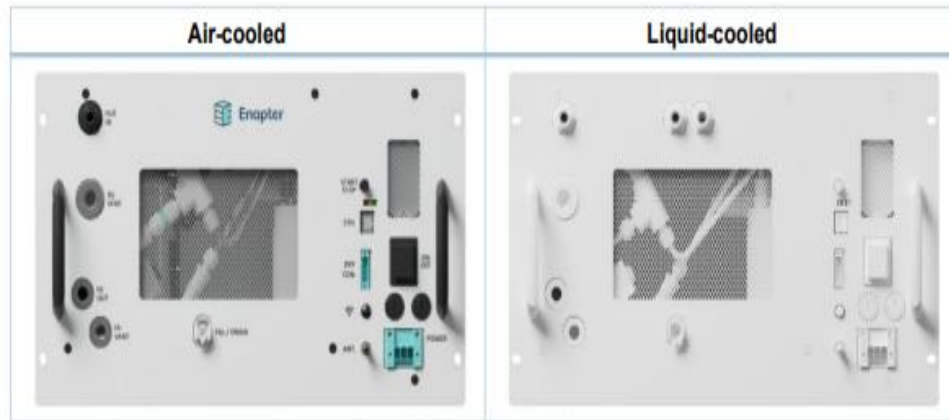
- **Diesel genset:**

- Model : Cummins 120 kW / 150 kVA Generator
- Peak power : 120 kW
- Efficiency = 45%
- Life span = 25 Years



2. System component sizing and Selection

- PtoG Seasonal storage: (H₂ Core systems GmbH)



Electrolyser:

- Input Power : 120 kW
- Efficiency : 53.33%
- Hydrogen production rate : 0.0005 kg/s or 64 kW
- Life span : 20 Year

Fuel cell:

- Output Power : 120 kW
- Efficiency : 43%
- Hydrogen supply rate : 0.0024 kg/s or ~ 279 kW
- Life span : 20 Year



2. System component sizing and Selection

- **Heat pump :** Thermal power rating = 80 kW, COP = 2.7, Type of Heat pump : Ground water, Life span = 25 Years (No specific provider)[1]

- **Gas Boiler:**

- Model :Bosch GC7000WP 150 23
- Peak Power capacity : 150 kW
- Efficiency = 94%
- Life span = 25 Years



- **Thermal storage:**

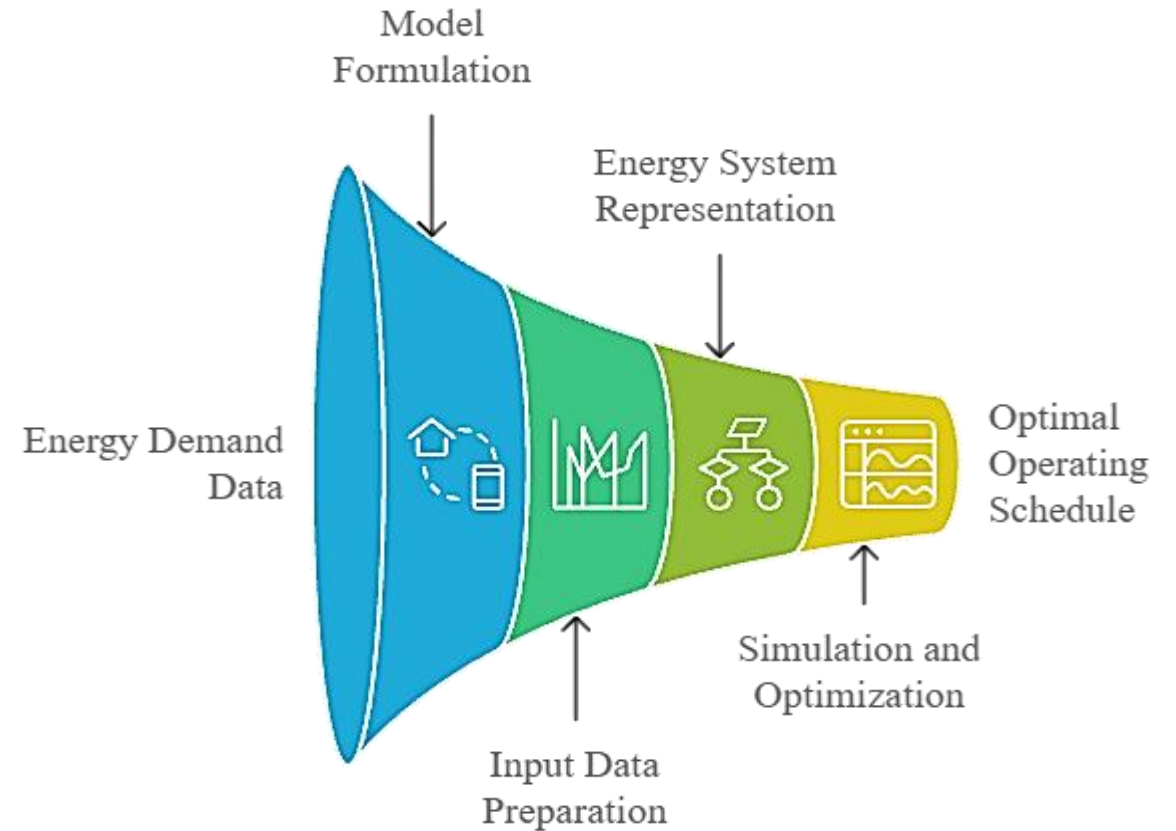
- Model : Sand battery
- Storage capacity : 60000 kWh
- Charging/Discharging Efficiency = 90%
- Self discharge rate = 0.3% per day
- Life span = 25 Years



- <https://www.google.com/acik?sa=L&ai=DChsSEwj19Kv687qSAX> accessed on 02.02.2026
- <https://polarnightenergy.com/wp-content/uploads/Lahti-Sand-battery-2-MW-1536x960-jpg.webp> accessed on 02.02.2026

3. Simulation and Optimization of Energy model

- **Modelling approach**
- **Input data preparation**
- **Energy System representation in Python**
 - Buses
 - Source and Sinks
 - Converters
 - Storages
- **Objective function and optimisation logic**
- **Post processing and validation checks**



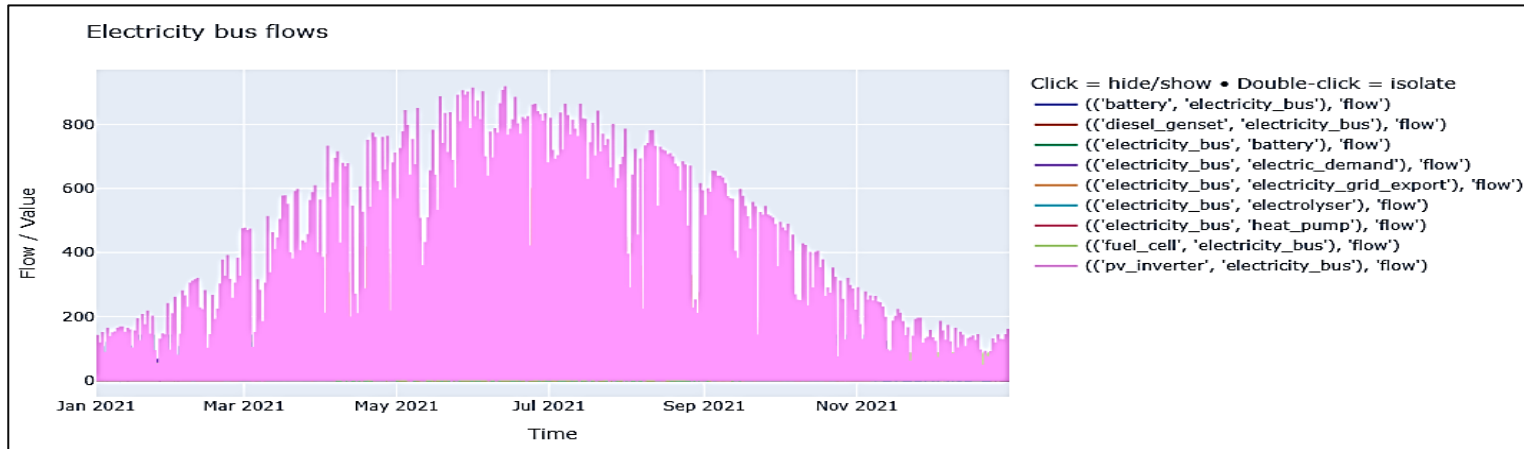
4. Results and Analysis

Optimization Results

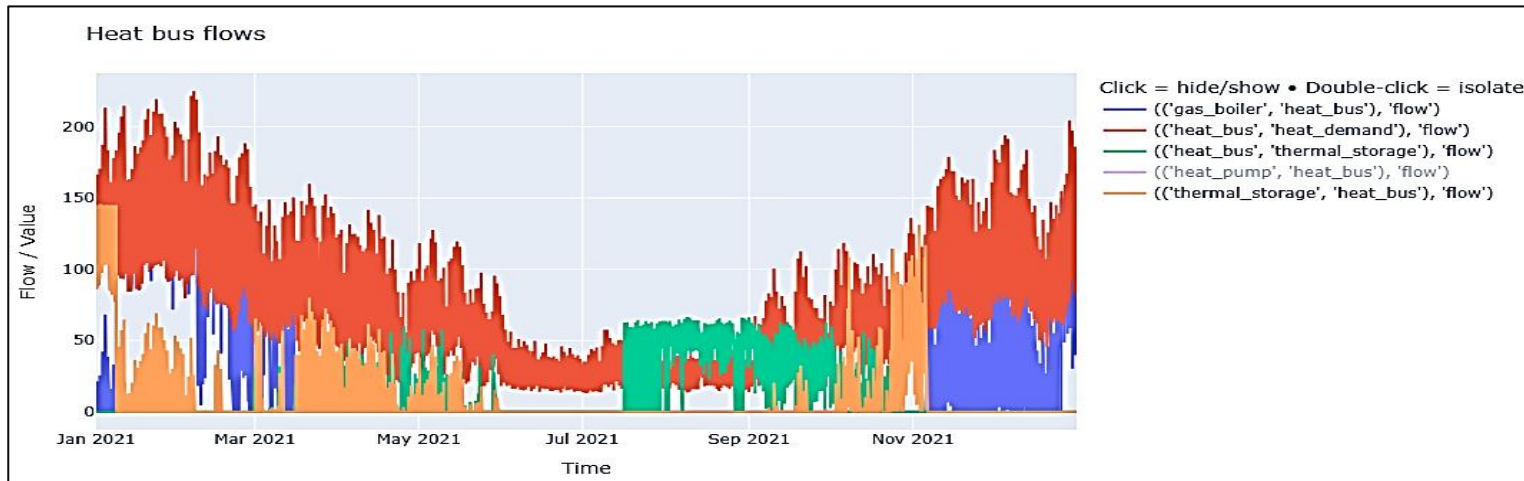
- Total energy produced by the PV is 1187507 kWh with max. DC power output of 937 kW.
- PV curtailment has been avoided by exporting electricity to grid.
- Stored Hydrogen is around 5000 kg, so that storage volume kept to 5100 kg.
- System sizes has been modified as per optimization result.
- Operating hours:
 - ❑ Heat pump = 7136 hours
 - ❑ Electrolyser = 3091 hours
 - ❑ Natural Gas Boiler = 2371 hours
 - ❑ Fuel cell = 1463 hours
 - ❑ Diesel Genset = 0 hours

4. Results and Analysis

- Electricity Bus Flow

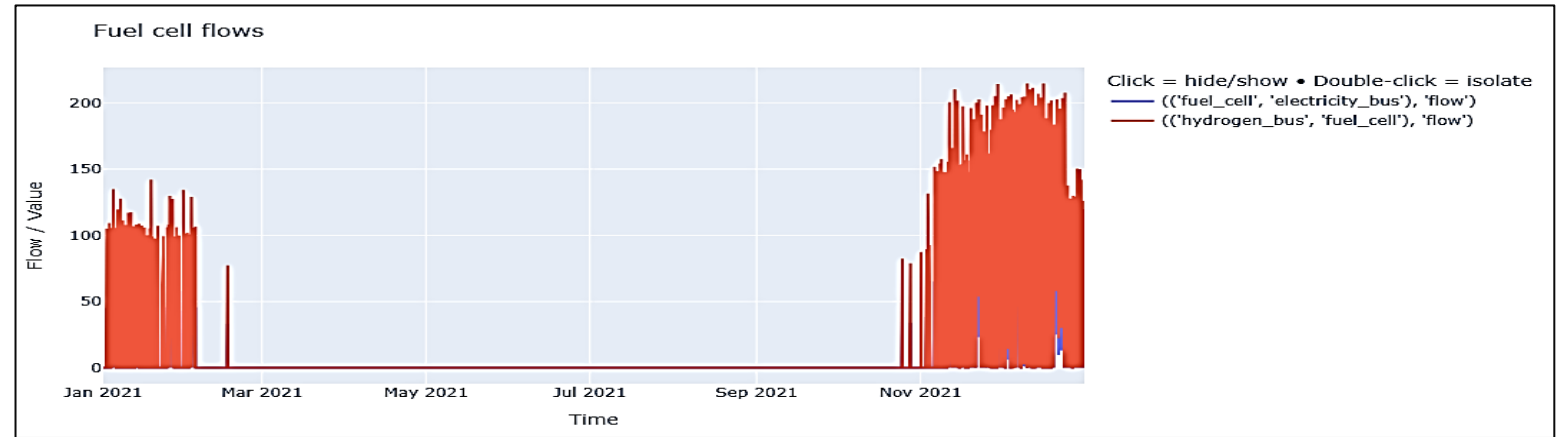


- Heat Bus Flow

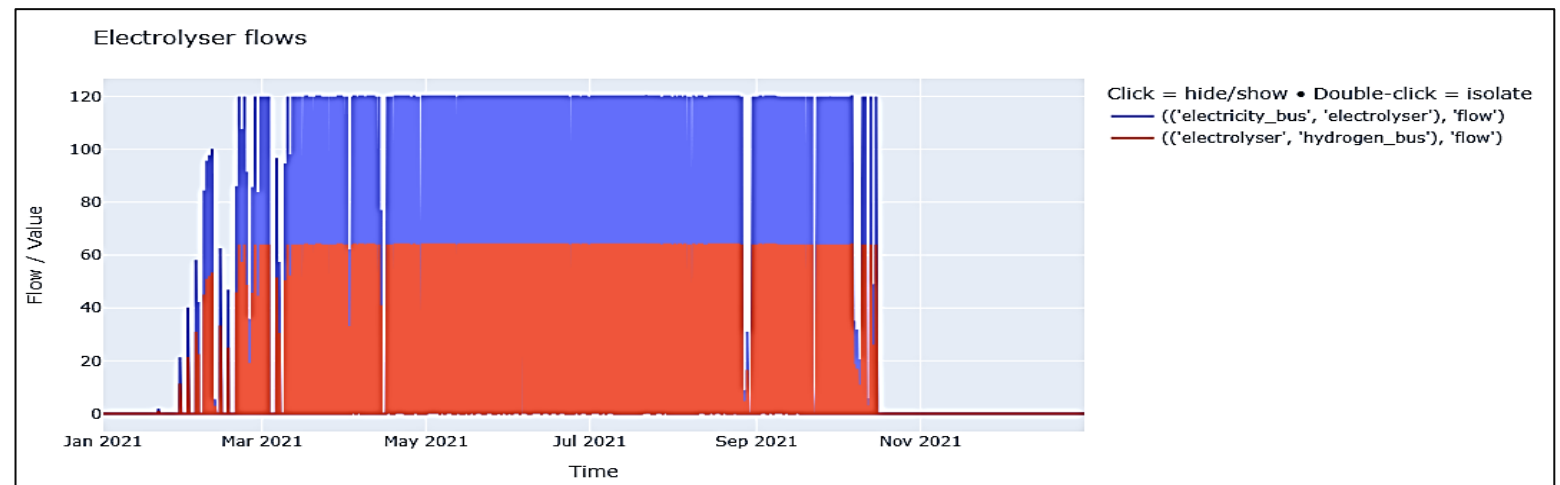


4. Results and Analysis

- Fuel Cell Flow

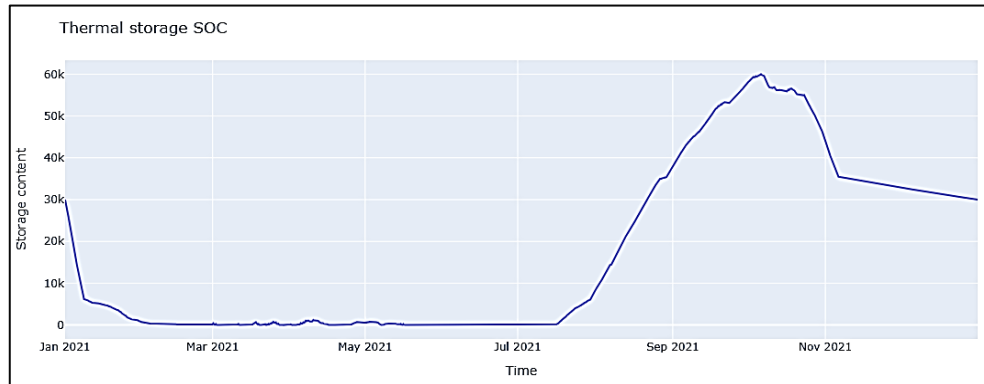


- Electrolyser Flow

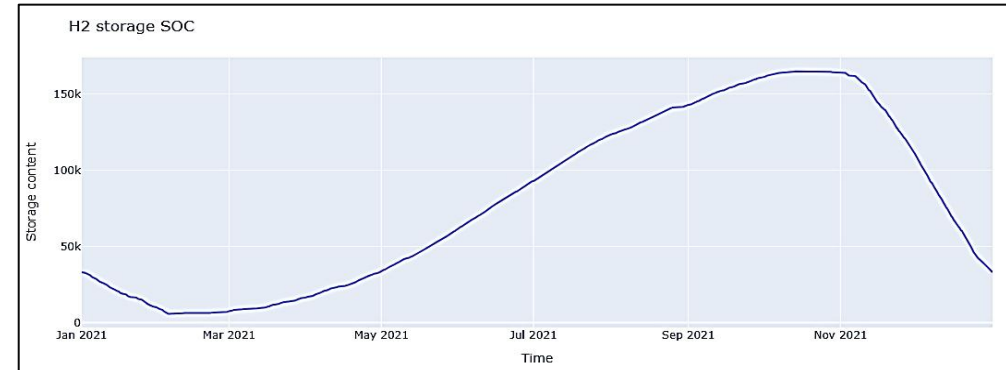


4. Results and Analysis

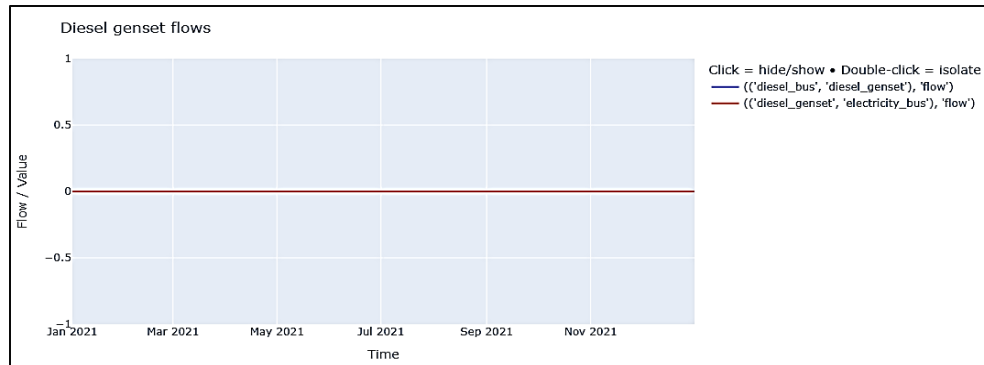
- Thermal Storage SOC



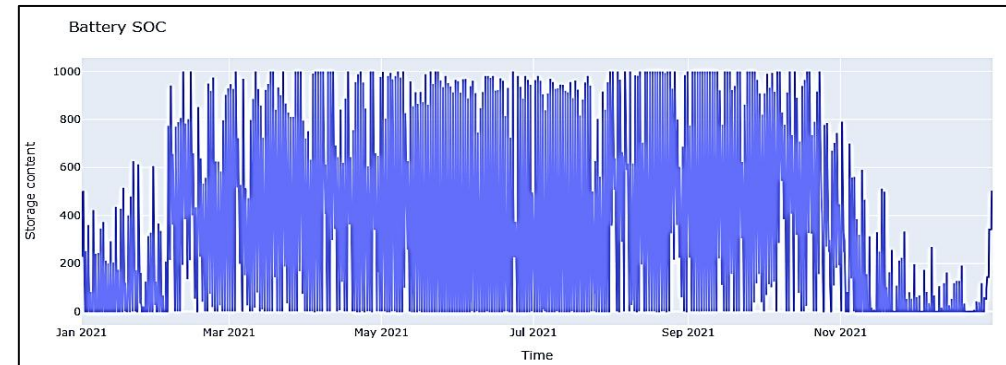
- H₂ Storage SOC



- Diesel Genset



- Battery Storage SOC



5. Economical calculation

Component	Cost of component	Lifespan (Years)	Total Investment cost (€)	Subsidy	Cost after subsidy
PV modules	664 €/kW	25	976080	0	976080
Inverter	47 €/kW	10	44650	0	44650
Battery storage	200 €/kWh	15	200000	0	200000
Electrolyser	2290 €/kW	20	274800	0	274800
Fuel cell	1836.5 €/kW	20	183650	0	183650
Hydrogen storage	0.451 €/kWh	20	74415	0	74415
Heat Pump	4000 €/kW	25	320000	55% KfW	144000
Gas Boiler	74 €/kW	20	11100	0	11100
Thermal storage	8 €/kWh	25	480000	55% KfW + 2500€ (Häuser Programm, Bavaria)	213500
Diesel genset	-	25	11000	0	11000
Total Capital Investment					2133195

Table 1: Capital Investment cost

5. Economical calculation

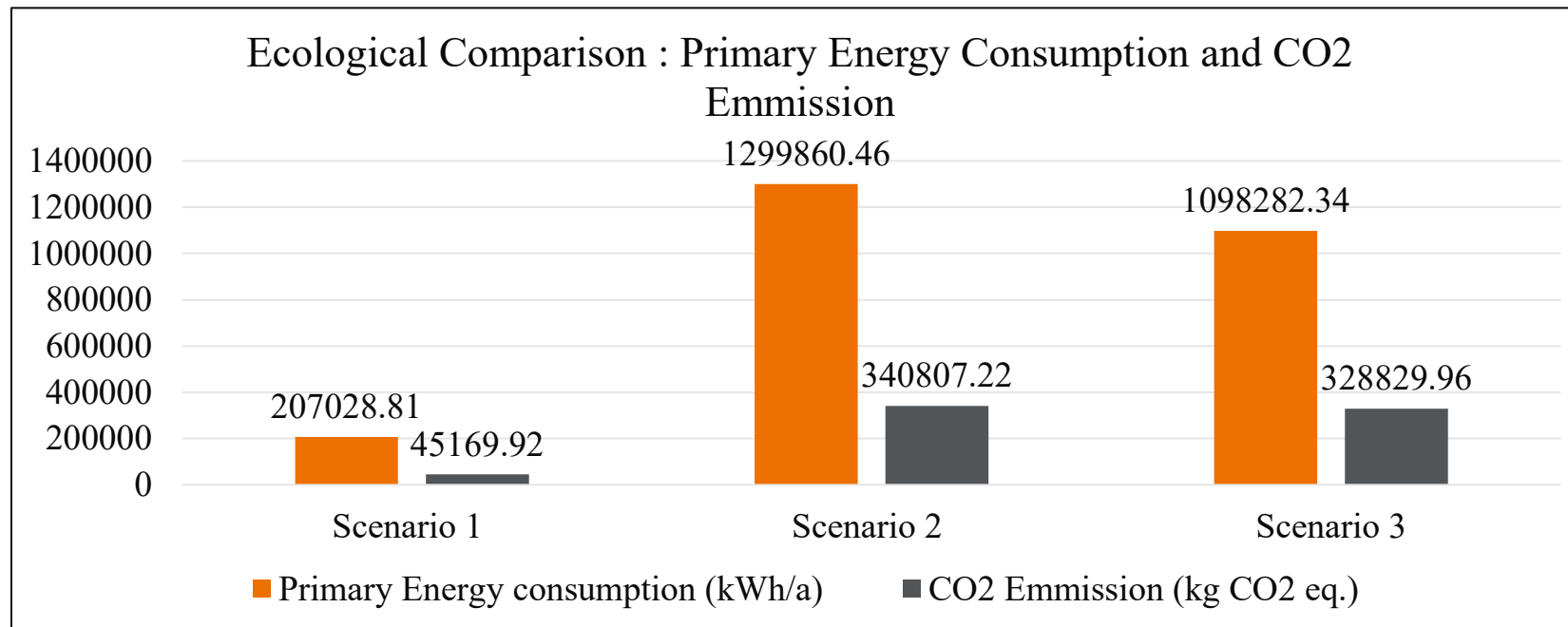
- Annual Interest rate : 6%
- Cost of Diesel : 0.0822 €/kWh
- Cost of natural gas : 0.12 €/kWh
- Feed in tarif : 0.0672 €/kWh
- Annual maintenance cost = 2% CAPEX
- Electricity demand = 341964.31 kWh/a
- Heat Demand = 584786.56 kWh/a
- LCOE = 26.29 Cents/kWh or 262.9 €/MWh
- $LCOE_{\text{Electricity}} = 63.91$ Cents/kWh
- $LCOE_{\text{Heating}} = 8.86$ Cents/kWh

Component	Annualised capex (€)
PV modules	102357.63
Inverter	6066.5
Battery storage	20592.55
Electrolyser	23958.32
Fuel cell	16011.44
Hydrogen storage	6684.44
Heat Pump	11264.65
Gas Boiler	967.75
Thermal storage	16701.4
Diesel genset	860.49
Maintenance cost	49356.88
Fuel cost	15470.7
Export Grid maintenance cost	0
Grid revenue	-26692.38
Total	243600.37

Table 2: Annualised Investment cost

6. Ecological Comparison

- Scenario 1 : Energy model
- Scenario 2 : Electricity - Grid , Heat - Natural gas Boiler
- Scenario 3 : Electricity - Grid , Heat – Heat Pump and Natural gas Boiler



7. Conclusion and Recommendation

7.1 Conclusion

- Solar energy has potential to fulfill energy demand if proper energy storage has been designed
- Use of daily energy storage leads to more utilization of solar energy
- PtoG seasonal energy storage has a lower round trip efficiency around 22.24%
- Heat pump is better solution for heating
- Use of thermal storage leads to lower operating hours of Natural gas boiler
- Reduction in CO₂ emission by almost 87% compared to rest scenarios

7. Conclusion and Recommendation

7.2 Recommendation

- Replace Natural gas boiler with Heat pump or electric boiler for peak heat demand for reduction in CO₂ emission.
- For reduction in operating hours of natural gas boiler, increase the size of PtoG components.
- Replacement of PtoG seasonal storage with another energy storage such as battery storage, Pump Hydro storage or Thermal storage leads to higher overall efficiency and lower investment cost.
- In case of grid operating charge, PV curtailment is better option by reducing the size of Inverter which again leads to decrease in LCOE.
- PV capacity also can be reduced if battery storage is used for seasonal storage, then PtoG seasonal storage

8. References

1. C. P. Underwood, “Heat pump modelling,” in *Advances in Ground-Source Heat Pump Systems*, S. J. Rees, Ed., Woodhead Publishing (Elsevier), Cambridge, UK, 2016, pp. 387–421. <https://doi.org/10.1016/B978-0-08-100311-4.00014-5>
2. H2 core systems GmbH, *Electrolyser and Fuel Cell Data*, Product catalogue, Heide, Germany, 2025
3. JA Solar, *JAM54D41LB n type Double glass Bifacial Modules*, Product catalogue, Beijing, China, 2025 <https://www.jasolar.com> last accessed on 29th Dec., 2025
4. GINLONG TECHNOLOGIES CO., LTD., *Solis Three Phase High Voltage Energy Storage Inverters*, Product catalogue, Ningbo, China, 2025 <https://www.solisinverters.com> last accessed 29th Dec., 2025
5. Polyntech, *Force H3 stackable battery*, Product catalogue, Shanghai, China, 2025, <https://en.pylontech.com.cn/products/forceh3> last accessed 29th Dec., 2025
6. Bosch Industrial Heat, *Condens 7000 WP*, Product catalogue, Worcester, UK, 2025 <https://www.bosch-industrial.com> last accessed 30th Dec., 2025
7. Cummins, *C170D5I*, Product catalogue, Columbus, U.S.A.- 2025 <https://www.cummins.com> last accessed 30th Dec., 2025
8. European commission, Photovoltaic Geographic Information System, <https://joint-research-centre.ec.europa.eu/photovoltaic-geogra> last accessed 15th Jan., 2026
9. EEG, “EEG subsidy and subsidy rates” EEG, 2026. <https://www.bundesnetzagentur.de> last accessed 23rd Jan., 2026

VIELEN DANK!

Group 02

Vasim Husenbhai Chudesara

Ashish Manoj Mistari

Jetson Joseph

Jesteeana Jose

